

Program : Diploma in Chemical Engineering	
Course Code : 3077	Course Title: Process Equipment Maintenance Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To provide basic knowledge about process equipments and piping system used in chemical process industries.
- To develop skills to maintain and operate valves, pumps, heat exchangers and piping system.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Unit operations		Chemical Process Principles	2

Course Outcomes :

On completion of the course student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Dismantle, identify the parts and assemble gate valve, globe valve, ball valve, plug valve, diaphragm valve and check valve.	18	Understanding
CO2	Dismantle, identify the parts and assemble centrifugal pump, reciprocating pump and centrifugal blower.	12	Understanding
CO3	Remove fouling in shell and tube heat exchanger.	3	Applying
CO4	Assemble a model piping by selecting appropriate pipe fittings.	12	Applying

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3		3					
CO4			3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Dismantle, identify the parts and assemble gate valve, globe valve, ball valve, plug valve, diaphragm valve and check valve.		
M1.01	Dismantle, identify the parts and assemble gate valve	3	Understanding
M1.02	Dismantle, identify the parts and assemble globe valve	3	Understanding
M1.03	Dismantle, identify the parts and assemble ball valve	3	Understanding
M1.04	Dismantle, identify the parts and assemble Plug valve	3	Understanding
M1.05	Dismantle, identify the parts and assemble Diaphragm valve	3	Understanding
M1.06	Dismantle, identify the parts and assemble Check valve	3	Understanding
CO2	Dismantle, identify the parts and assemble centrifugal pump, reciprocating pump and centrifugal blower.		
M2.01	Dismantle, identify the parts and assemble centrifugal pump.	3	Understanding
M2.02	Dismantle, identify the parts and assemble multi- stage pump.	3	Understanding
M2.03	Dismantle, identify the parts and assemble reciprocating pump	3	Understanding
M2.04	Dismantle, identify the parts and assemble centrifugal blower.	3	Understanding
	Series Test – I		

CO3	Remove fouling in shell and tube heat exchanger.		
M3.01	Dismantle, identify the parts and remove fouling in heat exchangers.	3	Applying
CO4	Assemble a model piping by selecting appropriate pipe fittings.		
M4.01	Identify the pipe fittings, flanges, elbow, tee, reducer, bend, spectacle blind, End flange.	3	Understanding
M4.02	Select suitable pipe fittings and make a model piping with 1” pipe.	3	Applying
M4.03	Select suitable pipe fittings and make a model piping with 2” pipe and identify the location for low point drain and high point vent.	3	Applying
M4.04	Open ended experiments	3	Applying
	Series Test – II		

Text / Reference

T/R	Book Title/Author
T1	Unit Operation – I; K A Ghavavni
R1	Mass transfer Operations; Warren L McCabe and Julian C Smith
R2	Introduction to Chemical Engineering; Walter L Badger and Julius T Bancherro